

Fleet replacement decisions under demand and fuel price uncertainties



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ABSTRACT

This paper proposes a real option model to investigate the fleet replacement timing decisions when a shipowner faces uncertain demand on various routes and uncertain fuel prices (both MGO and LNG prices). A multi-option least squares Monte Carlo simulation algorithm is utilized to find the replacement probabilities in future years and the expected NPVs of the cost savings after the replacement. The proposed model is applied to a chemical tanker shipping company. The case results indicate that the shipowner should replace all of his diesel-fuelled vessels with dual-fuelled vessels at one time, rather than replacing them one-by-one. Moreover, we compare the performance of two possible government policies, the vessel subsidy policy and the fuel subsidy policy. The sensitivity analysis indicates that in our case, the shipowner prefers the vessel subsidy policy, which brings him more cost-savings and allows him to make the earlier replacement decision. Alternately, the fuel subsidy policy is suitable for routes with a larger turnover volume if governments wish to see the more obvious policy feedbacks.

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1. Introduction

Shipowners currently face strict emission control. To comply with the 0.1% sulfur limit in the emission control areas (ECAs) after 2015, three alternatives can be used: switching to low sulfur fuel, installing an exhaust gas scrubber or installing LNG-compatible machinery (Acciario, 2014b; Patricksson et al., 2015). Among them, the use of LNG-compatible vessels has recently become more promising due to the decrease of LNG prices. However, this plan has a significant upfront cost, and its benefit heavily depends on the uncertain price advantages of LNG compared to marine gas oils (MGO). Alternately, uncertain future shipping demand is another “X factor” that affects the use of new energy vessels. Therefore, whether and when to replace their diesel-fuelled fleet with the new energy vessels are the crucial problems encountered by shipowners.

In this paper, we use a real option approach to study the fleet replacement timing decisions when a shipowner faces uncertain demand on various routes and volatile fuel prices (both MGO and LNG prices). We try to provide shipowners with management insights and governments with policy implications when they face dynamically uncertain contexts. Specifically, we aim to explore the following questions: (1) How can a shipowner's fleet replacement decisions be analyzed if the future demand on routes and fuel prices are uncertain and dynamically volatile? (2) How can the impacts of parameter changes, e.g., cargo demand and fuel price volatilities, on shipowner fleet replacement decisions be evaluated? (3) How can the performance of government policies (e.g., the subsidy for building new energy vessels or the subsidy for using LNG fuel) be compared under different scenarios? From the results of our model, not only can shipowners estimate their fleet replace-

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ment probabilities in future years and the expected fleet cost savings after replacement, but governments can also evaluate the impacts of their related policies.

Our work contributes to the fleet replacement problem under environmental compliance in the following two aspects.

First, our research can support shipowners in making appropriate decisions regarding fleet replacement, especially when they have heterogeneous vessels and face uncertain cargo demand on routes and volatile fuel prices. In a dynamically uncertain context, shipowners are confronted with the timing of their fleet replacement, i.e., either vessels should be replaced now or replacement should be deferred to the future. Moreover, it is more difficult for shipowners to make such decisions when they operate heterogeneous vessels because the benefits after the replacement are different for various types of vessels. Therefore, the replacement order has significant impacts on the final decisions. In our paper, the real option approach is used to investigate shipowners' tradeoff between obtaining the present cost savings and the deferral option value. The fleet replacement problem is transformed into an optimal stopping problem. Moreover, we take into account the option value of the replacement order, which is very crucial to the replacement decisions for heterogeneous vessels, and the multi-option least-squares Monte Carlo simulation (LSM) algorithm is developed to find the option value and the fleet replacement probabilities in future years for the best replacement plan.

Second, our sensitivity analysis based on a real chemical tanker shipping company can provide a sample for governments to investigate their policy impacts on shipowners' fleet replacement behavior. According to the policy practice proposed by the Ministry of Transport in China, we summarize two possible policies: the vessel subsidy policy and the fuel subsidy policy. We perform sensitivity analysis based on the wide ranges of the parameters. The results of our case study reveal that the shipowner prefers the vessel subsidy policy, which brings him more cost savings and enables him to make earlier replacement decisions. Alternately, the fuel subsidy policy is suitable for routes with a higher turnover volume,¹ if governments wish to see the more obvious policy feedback. Although our policy comparisons rely on a specific case study, the results can still provide useful policy implications to governments when they make policies to encourage shipowners to comply with stricter environmental regulations.

The rest of this paper is organized as follows: Section 2 provides a review on related literature. Section 3 describes the problem and proposes the model. In Section 4, we apply our model to a chemical tanker shipping company to investigate its fleet replacement decisions, and we analyze the impacts of different government policies. Conclusions and possible directions for future research are presented in Section 5.

2. Literature review

The related literature can be categorized into three main streams: literature on fleet deployment and replacement (or renewal), energy-efficient maritime transportation, and real option analysis in shipping.

There is much literature on the traditional fleet deployment and replacement (or renewal) problem. Some survey papers (Christiansen et al., 2013; Pantuso et al., 2014) provide comprehensive reviews on this issue. However, most studies investigate this problem in a deterministic setting. For the context of uncertainty, some papers address stochastic cases by replacing the uncertain data with its average or extreme values (e.g., Jaikumar and Solomon, 1987; Crary et al., 2002). Only a few papers explicitly consider uncertainty in their models, using stochastic programming (Meng and Wang, 2010; Meng et al., 2012; Schinas and Stefanakos, 2012; Dong et al., 2015; Pantuso et al., forthcoming), robust optimization (Alvarez et al., 2011), and simulation (Fagerholt et al., 2010; Shyshou et al., 2010; Halvorsen-Weare and Fagerholt, 2011; Song et al., 2015). Our paper differs from these studies in the following ways. In our paper, we consider a fleet replacement problem when the demand on routes and the fuel prices are uncertain, especially with nonstationary stochastic processes. Therefore, a shipowner's fleet replacement decisions are subject to changes year-by-year. In other words, the value of decision flexibility under uncertainty should be measured. We analyze the option to defer the fleet replacement and assess the value of postponing the replacement decisions. Moreover, we investigate the fleet replacement problem under environmental compliance and discuss the impacts of different government policies on shipowners' fleet replacement outcomes.

Energy-efficient maritime transportation studies have attracted much attention in recent years. Psaraftis and Kontovas (2013) have provided an excellent survey on the speed optimization models for energy-efficient maritime transportation. In addition, Schinas and Stefanakos (2012) used a linear stochastic programming model to estimate the cost impact of the sulfur limits on the increase of operating expenses of ships. Fagerholt et al. (2015) investigated the impacts of stricter emission limits in ECAs on the speed and routing decisions. Patricksson et al. (2015) proposed a stochastic programming model to examine a shipping operator's fleet renewal problem with regional emission limits. Compared to these papers, our paper considers the value of postponing the fleet replacement decisions, as well as the value of reordering the fleet replacement sequences.

Real options analysis (ROA) is a powerful tool to address the flexibility in an investment decision (Dixit and Pindyck, 1994). There are many applications in ship investment (e.g., Andersen, 1992; Goncalves, 1992; Dixit and Pindyck, 1994; Bendal and Stend, 2005; Sødal et al., 2008, 2009; Gkochari, 2015). However, they do not take into account the environmental compliance and the impacts of related government policies on shipowners' investment decisions.

¹ Turnover = shipping volume * voyage distance.

Next, we focus our attention on the work of Acciario (2014a,b), who has addressed investment timing decisions in LNG-fuelled vessels to comply with the upcoming environmental regulations using the real options approach. Although Acciario's topic and main approach are similar to our work, there are still various differences as follows: (1) Different methods. Although real options are used in both Acciario's work and ours, a discrete binomial tree approach is selected by Acciario (2014a,b), while a multi-option least-squares Monte Carlo simulation is employed in our paper to solve the optimal fleet replacement problem. Models proposed by Acciario only describe the fuel scenarios where the highest and the lowest possible amounts of savings (or losses) are calculated, whereas our model can examine more scenarios by assuming that the demand on routes follows the geometric Brownian motion and that the fuel prices follow mean-reverting processes. (2) Different uncertain items. Acciario (2014a,b) considers the fuel price uncertainty, while we take into account both the demand on routes and the fuel price uncertainties. We consider three stochastic processes, which makes the problem much more complicated than in Acciario (2014a,b). Moreover, as we have illustrated in the case study, the impacts of the small volatility in demand may be magnified by the relatively larger volatilities of the fuel prices. Therefore, our work can solve more problems in reality. (3) Different scales. Models used by Acciario (2014a,b) are applied to a handysize vessel, while our model addresses a company's fleet replacement with heterogeneous vessel types serving various shipping routes. In addition to the vessel replacement time, our model can obtain richer results, e.g., a shipowner can know the number of vessels and voyages deployed, the number of vessels chartered in, the number of his own vessels chartered out and the lay-up days. These decisions are related to the fleet operation costs, which finally affect the optimal replacement timing. Therefore, our work is suitable for large shipowners who operate fleets with heterogeneous vessels serving various shipping routes.

3. Problem description and mathematical models

In this section, the problem is first described. Then, a fleet deployment model is introduced to examine the fleet replacement decision under certainty. Following that, the geometric Brownian motion (GBM) and the mean-reverting process are used to characterize the dynamic evolutions of the demand on routes, as well as the prices of different fuels, i.e., MGO and LNG prices. Next, a real option model is developed to investigate a shipowner's fleet replacement timing decisions under uncertainty. Finally, we explain how to use our model to analyze the policy impacts on the shipowner's decisions.

3.1. Problem description

In our problem, a shipping company operates a heterogeneous fleet of ships with different sizes. The shipowner needs to decide the annual fleet deployment plan to fulfill her shipping business at several routes. Currently, all of her vessels use diesel engines. Following the enforcement of stricter ECA regulations on sulfur and nitrogen oxides (NO_x), shipowners are considering the possibility of replacing conventional diesel-fuelled vessels with LNG-powered ones. LNG-powered vessels have higher acquisition costs but possibly lower fuel costs. Therefore, the shipowner should compare the annual costs before and after the replacement to make the decision. However, in today's volatile shipping market, demand on routes and fuel prices (including MGO and LNG prices) are uncertain in the future. The shipowner faces a series of replacement timing decisions where he should consider the tradeoff between making a replacement immediately and waiting for better opportunities in the future. Such replacements have many different plans according to different orders (i.e., what fleet with a specific size should be replaced firstly). The shipowner should choose the best plan to maximize her expected total cost savings within a future period.

3.2. Fleet deployment model

In this section, we use the fleet deployment model introduced by Gelareh and Meng (2010) to examine a shipowner's short-term vessel planning. A shipping company operates a heterogeneous fleet of diesel-fuelled ships with different sizes. The set of ship types is denoted by V . N_v is the number of ships of type v . Each ship type has a cargo carrying capacity K_v . N_v^I is the number of ships of type v available in the market for chartering in. C_v^I and C_v^O are the costs of chartering in and out a ship of type v for one year, respectively. C_v^L is the lay-up cost for a ship of type v . The set of predefined liner routes is denoted by R . D_r is the demand on route r for one year. F_r is the frequency requirement for route r in one year. C_{vr} and T_{vr} are the voyage cost and voyage time on route r with ship type v , respectively. T^H is the length of the planning period, i.e., 365 days.

The decision variables are defined as follows. x_{vr} is the number of voyages performed by ship type v on route r . n_{vr} is the number of ships of type v deployed on route r . n_{vr}^I is the number of ships of type v chartered in and deployed on route r . n_v is the number of purchased ships of type v . n_v^O is the number of owned ships of type v chartered out. d_v is the number of lay-up days for ships of type v . We use TC to represent the operational cost for a fleet deployment plan.

Using the above symbols, we express a shipowner's fleet deployment model as follows:

$$TC = \min \sum_{v \in V} \sum_{r \in R} (C_{vr} x_{vr} + C_v^l n_{vr}^l) + \sum_{v \in V} (C_v^l d_v - C_v^o n_v^o) \quad (1a)$$

$$\text{s.t.} \quad \sum_{v \in V} K_v x_{vr} \geq D_r, \quad \forall r \in R \quad (1b)$$

$$\sum_{v \in V} x_{vr} \geq F_r, \quad \forall r \in R \quad (1c)$$

$$\sum_{r \in R} n_{vr} \leq N_v, \quad v \in V \quad (1d)$$

$$\sum_{r \in R} n_{vr}^l \leq N_v^l, \quad v \in V \quad (1e)$$

$$n_v^o \leq N_v - \sum_{r \in R} n_{vr}, \quad v \in V \quad (1f)$$

$$T_{vr} x_{vr} \leq T^H (n_{vr} + n_{vr}^l), \quad v \in V, \quad r \in R \quad (1g)$$

$$d_v = T^H N_v + \sum_{r \in R} T^H n_{vr}^l - T^H n_v^o - \sum_{r \in R} T_{vr} x_{vr}, \quad v \in V \quad (1h)$$

$$x_{vr}, n_{vr}, n_{vr}^l \in \{\mathbb{Z}^+ \cup 0\}, \quad v \in V, \quad r \in R \quad (1i)$$

$$n_v^o \in \{\mathbb{Z}^+ \cup 0\}, \quad v \in V \quad (1j)$$

The objective function (1a) aims to minimize the fleet operation costs, which consist of the sailing costs ($C_{vr} x_{vr}$), the chartering costs ($C_v^l n_{vr}^l$), and the lay-up costs ($C_v^l d_v$), minus the revenue from chartering out ships ($C_v^o n_v^o$). Constraints (1b) ensure that the shipping demand is satisfied for every route. Constraints (1c) guarantee that the frequency requirement for every route is met. Constraints (1d) and (1e) require that the owned and chartered-in vessels of each type used on all routes do not exceed the available vessels. Constraints (1f) represent the unused ships that can be chartered out. Constraints (1g) ensure that the total voyage time on each route by each ship type is limited by the number of owned and the chartered-in vessels. Constraints (1h) define the lay-up time for each ship type. Constraints (1i) and (1j) guarantee the non-negativity and the integer requirements for the decision variables.²

In model (1a)–(1j), the voyage cost C_{vr} is determined as follows if the fleet is composed of the diesel-fuelled vessels:

$$C_{vr} = (T_{vr} \cdot OCS_v + BD_{vr} \cdot OCB_v) \cdot F_O + (T_{vr} + BD_{vr}) \cdot DFC_v \quad (2)$$

where the voyage time $T_{vr} = \text{Dis}_r / (24 \cdot Sp_v)$, Dis_r is the voyage distance of route r (miles), Sp_v the sailing speed of a ship of type v (miles/hour), OCS_v and OCB_v the MGO consumption rates of a ship of type v when sailing and in berth (tons/day), respectively, BD_{vr} the days in berth for a ship of type v on route r , F_O the MGO price (\$/ton), and DFC_v the daily fixed cost of a ship of type v (\$/day).

The LNG-propelled ships in operation are either equipped with lean burn gas engines or dual-fuel engines. The gas engines are based on spark plug ignition. Dual-fuel engines can run in either gas mode or diesel mode. In gas mode, the engine works according to the lean burn Otto principle, but the lean air mixture is ignited by the injection of a small amount of diesel fuel, i.e., pilot fuel, into the combustion chamber instead of a spark plug (Doug, 2010). In our paper, an LNG-powered vessel refers to one equipped with dual-fuel engines because the only figures we could acquire related to ships with dual-fuel engines. If the fleet is composed of dual-fuelled vessels, its voyage cost C_{vr} can be expressed as follows:

$$C_{vr} = T_{vr} \cdot GCS_v \cdot F_G + BD_{vr} \cdot OCB_v \cdot F_O + (T_{vr} + BD_{vr}) \cdot DFC_v \quad (3)$$

where GCS_v is the LNG consumption rate of a dual-fuelled ship of type v when sailing (tons/day), and F_G is the LNG price (\$/ton).

In model (1a)–(1j), we do not consider the freight rate. We use the cost minimization approach to examine the fleet deployment decision. The reasons are as follows. In most cases, a shipowner does not have enough market power to affect the freight rate and behaves as a price-taker. Therefore, the results of the cost minimization equal the profit maximization given the fixed freight rates. Moreover, the cost minimization approach is a commonly used method in the fleet replacement literature (Pantuso et al., 2014).

However, shipping speed is not a decision variable in our model, which means that a shipowner has to charter the vessels in/out or lay up the surplus vessels, rather than adjusting voyage numbers by changing the shipping speed, when facing a demand fluctuation. The main reason not to consider the speed decision is to simplify the problem because the relation between the speed and the costs is nonlinear. When there is a minor demand increase, a shipowner will have to put one more ship in the active fleet if he faces a contract constraint. That is, if the ship owner signs long-term contracts with ship-

² In some cases, ships are deployed on different routes that are rather independent and have different market demands. Some may think that it is sufficient to consider a one-ship-at-a-time replacement decision. However, a shipowner may need to deploy more than one ship on a route if it has a large demand. Besides the number of vessels deployed, he needs to make other decisions, e.g., the number of voyages, the number of vessels chartered in (if the demand is higher than expected), the number of his own vessels chartered out or the lay-up days (if the real demand is low). Note that these decisions are related to the fleet operation costs TC , which finally affects the optimal replacement timing. Therefore, the shipowner should consider a fleet (with many homogenous vessels) replacement decision in the case of route independence. Our model, (1a)–(1j), can cover this case by specifying that $v = 1$.

pers to transport large amounts of cargo (i.e., Contract of Affreightment), or he needs to transport cargo to meet his own trade demand, he should fulfill the shipping tasks regardless of whether the earnings from the extra demand can offset the cost. If the demand is low and the market freight rate is also low, a shipowner faces two choices: chartering his surplus vessels out or laying them up. His decisions depend on the comparison between the lay-up costs C_v^L and the costs of chartering out C_v^O . If it is difficult to charter them out, the shipowner will use the lay-up strategy. In our model, if C_v^O is small enough (the charter rate is very low), the optimal n_v^O (the number of owned ships chartered out) will be 0, and d_v (the lay-up days for ships) is positive.

3.3. Fleet replacement decisions under certainty

When facing a fleet replacement decision, a shipowner should compare the costs before and after the replacement. Here, we define the annual cost savings after the fleet replacement as follows:

$$\Delta TC = TC - TC' - \sum_v (P_{LNG} - P_{MGO}) \cdot AF \quad (4)$$

where ΔTC is the annual cost savings, TC and TC' are the annual total fleet costs³ defined by the fleet deployment model (1a)–(1j) before and after the fleet replacement, respectively, P_{LNG} and P_{MGO} are the prices of a dual-fuelled ship and a diesel-fuelled ship, respectively, and AF is the annuity factor.⁴

Eq. (4) means that the annual cost savings of the fleet replacement equal the annual fleet operational cost savings minus the annuity of the price difference between the dual-fuelled fleet and the diesel-fuelled fleet. For example, a company now deploys only diesel-fuelled vessels with different sizes, including 13,000-DWT vessels and 19,000-DWT vessels. If the shipowner wants to replace the 19,000-DWT diesel-fuelled vessels with dual-fuelled vessels (19,000 DWT), he should consider the operational cost savings for the original fleet deployment plan (the fleet composed of the diesel-fuelled vessels, including the 13,000-DWT vessels and the 19,000-DWT vessels) and the suggested fleet deployment plan (the fleet composed of the 13,000-DWT diesel-fuelled vessels and the dual-fuelled vessels), as well as the extra capital expenses for purchasing the dual-fuelled vessels. If the annual operational cost savings are larger than the annuity of the additional vessel investments, the replacement decision is feasible.

Such a decision is easy to make when the demand on routes and the MGO and LNG prices are certain. However, the demand on routes and the fuel prices always demonstrate strong fluctuations, especially in today's volatile world economy and trade environment. Next, we focus our attention on fleet replacement decisions under uncertain demand on routes and fuel prices.

3.4. Fleet replacement decisions under uncertainty

When facing uncertainties, a shipowner's fleet replacement decisions become difficult because he needs to consider the tradeoff between making a replacement immediately and waiting for better opportunities in the future. In other words, he should measure the value of decision flexibility under uncertainty. Real option analysis is an effective approach to address investment decisions in high capital cost industries with high volatilities (Dixit and Pindyck, 1994). To incorporate the uncertainty impacts into the investment decision, the traditional NPV rule is expanded as follows (Trigeorgis, 1996):

$$\text{Expanded NPV} = \text{NPV} + \text{option premium} \quad (5)$$

where the NPV is expanded by incorporating the “option premium” to represent the value of options.

In this section, we first describe the characteristics of the demand and the fuel prices. Then, we propose an option model for the fleet replacement decisions under uncertainty. Finally, we discuss the solution algorithm.

3.4.1. Uncertain demand and prices of MGO and LNG

To describe the characteristics of the uncertain demand on routes, we assume that it follows a GBM expressed as follows (Chow and Regan, 2011a,b):

$$dD_r = \alpha_r D_r dt + \sigma_r D_r dW, \quad \forall r \in R \quad (6)$$

where α_r is the mean-drift of the demand D_r , σ_r is the volatility of demand D_r , dW is an increment of a standard Wiener Process, and $dW \sim N(0, dt)$.

For the fuel prices, we assume that the logarithms of the MGO price and the LNG price follow the mean-reverting processes as follows (Dixit and Pindyck, 1994):

$$dF_0 = \eta_0 (\bar{F}_0 - F_0) dt + \sigma_0 dW \quad (7)$$

³ Note that, in the fleet deployment model (1a)–(1j), the length of the planning period is 365 days. Therefore, the total cost TC is an annual cost.

⁴ In this paper, the annuity factor is the sum of the discount factors for the first year through the last year of a vessel's life (20 years), inclusive, when the cost of capital is the same for all relevant years. $AF = \frac{1}{i} - \frac{1}{i(1+i)^n}$, where i is the discount rate and n is the vessel life.

$$dF_G = \eta_G(\bar{F}_G - F_G)dt + \sigma_G dW \quad (8)$$

where η_0 and η_G are the speed parameters of reversion, $\bar{F}_0$ and $\bar{F}_G$ the mean logarithm levels of the MGO price and the LNG price, respectively, σ_0 and σ_G the volatility parameters.

For simplicity, we assume that the demands of different routes are independent and that they are independent of the MGO and LNG prices, too.

3.4.2. Option model for replacement decisions under uncertainty

Unlike a one-shot investment decision, our fleet replacement problem requires ordered sequence decisions. That is, the payoffs of later decisions depend not only on themselves but also on whether the previous ones are exercised. Therefore, in our problem, the total option value of all the fleet replacement decisions consists of two parts: the maximum of all the options and the option to rearrange the replacement decisions. In the first part, each option treats the fleet replacements as purely compound options where one replacement decision depends on whether the prior replacement decision is exercised. The second part is the option to re-order the replacement decisions. For example, a shipowner now has a fleet composed of only the diesel-fuelled vessels, including the 13,000-DWT vessels and the 19,000-DWT vessels. A fleet replacement plan can be arranged in the following order: replace the 13,000-DWT diesel-fuelled vessels with the dual-fuelled vessels first, and then the 19,000-DWT diesel-fuelled vessels. In such a case, we treat the two replacement decisions as two options. The second replacement decision depends on whether the first one is exercised. In addition, the shipowner may have different replacement plans, e.g., replace the 19,000-DWT diesel-fuelled vessels with the dual-fuelled vessels first, and then the 13,000-DWT diesel-fuelled vessels. Note that this plan has a different replacement order from the previous one. Therefore, the total option value of the fleet replacement is the sum of these two parts, which can be expressed as follows:

$$\Phi = \sum_j NPV_j + \sum_j (F_{Dj} + F_{Sj}) + F_R \quad (9)$$

where the subscript j represents a replacement decision in a specific plan, Φ is the total option value of the fleet replacement, NPV_j the net present value of the cost savings after the j th replacement (i.e., the net present value of ΔTC after the j th replacement), F_{Dj} the deferral option value, F_{Sj} the option value of the subsequent replacements, and F_R the option value to rearrange the replacement decisions.

According to the framework proposed by Gamba (2002) and Chow and Regan (2011b), for each replacement plan p , the compound options can be modeled with the following Bellman equation:

$$\Phi_{p_j, t_n}(\mathbf{D}_{t_n}, F_{O, t_n}, F_{G, t_n}) = \max_{z_{p_j, t_n}} \left\{ \Delta TC_{p_j, t_n}(\mathbf{D}_{t_n}, F_{O, t_n}, F_{G, t_n}, \mathbf{z}_{p_j, t_n}) + \Phi_{p_{j+1}, t_n}(\mathbf{D}_{t_n}, F_{O, t_n}, F_{G, t_n}), (1 + \rho)^{-\Delta t} E \left[\Phi_{p_j, t_{n+1}}(\mathbf{D}_{t_{n+1}}, F_{O, t_{n+1}}, F_{G, t_{n+1}}) \right] \right\} \quad (10)$$

where the subscript p_j represents the j th decision in the replacement plan p , the subscript t_n is the time step, $\mathbf{D}_{t_n}$ is the routes demand vector at time step t_n , i.e., $\mathbf{D}_{t_n} = (D_{1, t_n}, D_{2, t_n}, \dots, D_{r, t_n})$, $\forall r \in R$, F_{O, t_n} and F_{G, t_n} the MGO price and the LNG price at time step t_n , respectively, $\mathbf{z}_{p_j, t_n}$ the decision variable vector at time step t_n for Problem (1a)–(1j), i.e., $\mathbf{z}_{p_j, t_n} = (x_{vr, t_n}, n_{vr, t_n}, n_{vr, t_n}^I, n_{vr, t_n}^O, d_{v, t_n})$, $\forall v \in V, \forall r \in R$, Φ_{p_j, t_n} the option value of the j th decision in Plan p at time step t_n , $\Delta TC_{p_j, t_n}$ the cost savings after the replacement in the j th decision in Plan p at time step t_n , and ρ the discount rate.

The Bellman Eq. (10) means that one tries to make decisions in each time state to maximize the outcome by either replacing his fleet immediately (exercising the option now) or deferring the replacement to the next time step (keeping the option). The payoff value of immediately making the j th replacement decision in Plan p is equal to the payoff from immediate fleet replacement of the j th decision plus the opportunity payoff of the $(j + 1)$ th decision. As noted by Chow and Regan (2011b), a backward option recursion is added to the backward time recursion in the multi-option cases. The total option value for Plan p at time step t_n is the sum of the option value of all its decisions, i.e., $\sum_j \Phi_{p_j, t_n}(\mathbf{D}_{t_n}, F_{O, t_n}, F_{G, t_n})$. If the shipowner has many plans, he should compare them and choose the plan that offers the highest first option in the sequence:

$$\Phi_{t_n}(\mathbf{D}_{t_n}, F_{O, t_n}, F_{G, t_n}, z_{t_n}) = \max_p \left\{ \sum_j \Phi_{p_j, t_n}(\mathbf{D}_{t_n}, F_{O, t_n}, F_{G, t_n}) : p \in \arg \max_p [\Phi_{p_1, t_n}(\mathbf{D}_{t_n}, F_{O, t_n}, F_{G, t_n})] \right\} \quad (11)$$

3.4.3. Solution algorithm

The Bellman Eq. (10) is difficult to solve. In this paper, we use the multi-option least-squares Monte Carlo simulation (LSM) algorithm to solve it. Gamba (2002) and Chow and Regan (2011b) indicate that the multi-option LSM algorithm is an extension of the LSM refined by Longstaff and Schwartz (2001). Stentoft (2004) proves that LSM is cost-effective compared to binominal lattice methods and Monte Carlo simulation and can converge towards the actual American option value, as the number of simulation paths, time steps and polynomials for regression approach infinity. Using their framework, we describe the solution algorithm of our problem as follows:

Step 1: For each simulation path $\omega \in \Omega$ and time t_n ,⁵ generate the simulated route demand using the following equation:

$$D_{r,t_n} = D_{r,t_{n-1}} \exp[(\alpha_r - \sigma_r^2/2)\Delta t + \sigma_r W_{\Delta t}], \quad \forall r \in R \quad (12)$$

Here, the Wiener Process $W_{\Delta t}$ can be simulated as $W_{\Delta t} = \varepsilon\sqrt{\Delta t}$, where $\varepsilon \sim N(0, 1)$.

MGO and LNG prices can be generated by the following equations (Dixit and Pindyck, 1994):

$$F_{O,t_n} = \bar{F}_O[1 - \exp(-\eta_O)] + F_{O,t_{n-1}} \exp(-\eta_O) + \frac{\sigma_O^2}{2\eta_O}[1 - \exp(-2\eta_O)]W_{\Delta t} \quad (13)$$

$$F_{G,t_n} = \bar{F}_G[1 - \exp(-\eta_G)] + F_{G,t_{n-1}} \exp(-\eta_G) + \frac{\sigma_G^2}{2\eta_G}[1 - \exp(-2\eta_G)]W_{\Delta t} \quad (14)$$

Step 2: For each Plan p , enumerate its every possible sequence, which is the all possible replacement arrangements in every year.⁶ For each sequence, use the simulated $D_{r,t_n}(\omega)$, $F_{O,t_n}(\omega)$ and $F_{G,t_n}(\omega)$ to solve Problem (1a)–(1j) and obtain $\Delta TC_{p_j,t_n}$ at each time step t_n and simulation path ω .

Step 3: Starting from the last time state, use the multi-option LSM to solve the Bellman Eq. (11). The detailed procedure of this step is as follows:

3.1 In the last time step t_n , let $\Phi_{p_j,t_n}(\omega) = \max(0, \Delta TC_{p_j,t_n})$.

3.2 Keep track of the optimal decision with the variable $\theta_{p_j}(\omega, t_n)$. If $\Delta TC_{p_j,t_n}(\omega) > 0$, then the sample path is “in the money” (Chow and Regan, 2011b) and $\theta_{p_j}(\omega, t_n) = 1$. Otherwise, $\theta_{p_j}(\omega, t_n) = 0$.

3.3 Let $n = n - 1$. If $n = 0$, go to Step 4;

3.4 Estimate $E(\Phi_{p_j,t_{n+1}})$ using the least squares regression with the Hermite polynomial series:

$$H_i(x) = (-1)^i e^{x^2/2} \frac{d^i}{dx^i} e^{-x^2/2} \quad (15)$$

Therefore, a regression function with N polynomials can be expressed as

$$E(\Phi_{p_j,t_{n+1}}) = \sum_{i=0}^N \beta_i (-1)^i e^{x^2/2} \frac{d^i}{dx^i} e^{-x^2/2} \quad (16)$$

where the coefficients β_i can be estimated using least squares. The variable x in (15) and (16) represents $\Delta TC_{p_j,t_n}(\omega)|_{\Delta TC_{p_j,t_n}(\omega) > 0}$ at each simulation path at the particular time state.

3.5 Compare the value of immediately replacing at the present time t_n with the expected value from future replacement (the discounted value of the estimated $E(\Phi_{p_j,t_{n+1}})$). If $(1 + \rho)^{-\Delta t} E[\Phi_{p_j,t_{n+1}}(\omega)] > \Delta TC_{p_j,t_n}(\omega) + \Phi_{p_j,t_n}(\omega)$, using the current fleet is better than the immediate replacement. Therefore, the current option and all subsequent options in the simulation path should be deferred, $\theta_{p_j}(\omega, t_n) = 0$. Otherwise, $\theta_{p_j}(\omega, t_n) = 1$.

3.6 If $n > 0$, go to Step 3.3.

Step 4: When $n = 0$, discount back the option value for each j th option and take the average:

$$E[\Phi_{p_j,t_0}(\omega)|\theta_{p_j}(\omega, t_n) = 1] = \frac{\sum_{\omega} \sum_n \Phi_{p_j,t_n}(\omega)(1 + \rho)^{-n}}{|\Omega|} \quad (17)$$

where $|\Omega|$ represents the number of simulation paths.

Repeat Step 2–Step 3 until the option values from all of the sequences of all of the plans are obtained.

Step 4: The option value is the maximum of the sum of option values obtained from the sequence belonging to the plan with the maximum first replacement, which is the solution of (11).

To show the correctness of our models and method, we provide a small example considering a replacement decision for one vessel in one route in Appendix A.

⁵ In our case, we set the time step as 1 year.

⁶ For example, a fleet replacement plan has the following order: replace the 13,000 DWT diesel fuelled vessels with the dual fuelled vessels first, then the 19,000 DWT diesel fuelled vessels. If we consider a three-year horizon, we have 3 possible replacement sequences: (i) replace the 13,000 DWT vessels at Year 1, then the 19,000 DWT vessels at Year 2 or 3; (ii) replace the 13,000 DWT vessels at Year 1 or 2, then the 19,000 DWT vessels at Year 3; (iii) replace the 13,000 DWT vessels at Year 2, then the 19,000 DWT vessels at Year 3.

3.5. How to use the models to analyze policy impacts

To encourage shipowners to use low-emission vessels, governments can use subsidy policies to affect shipowners' vessel choices. For example, the Ministry of Communication in China regulates a vessel subsidy policy whereby river carriers can receive a subsidy from the government if they use LNG-powered (or dual-fuelled) vessels according to the following criteria: RMB 630,000 per vessel with main engine power of 300 kw, RMB 780,000 per vessel with main engine power of 300–600 kw, RMB 900,000 per vessel with main engine power of 600–1000 kw, and RMB 1,000,000 per vessel with main engine power of 1000 kw. Although this subsidy policy is only for river ships now, it can be expanded to other ships. Another possible subsidy policy (in discussion now) is a fuel subsidy policy that gives shipowners lower LNG prices. Thus, it is natural to ask the following questions. Which policy do shipowners prefer? Are there any different impacts for these policies to shipowners' replacement timing decisions? What are the impacts of the policies to shipowners if the market changes (e.g., changes of demand volatilities on routes or changes of MGO or LNG price volatilities)?

To answer these questions, we use sensitivity analysis to evaluate the policies' performances. The vessel subsidy reduces the dual-fuelled ship price, whereas the fuel subsidy reduces the LNG prices in our models. Therefore, we will compare their impacts on the total option values and the fleet replacement probabilities every year for each replacement plan. Moreover, given the specific changes of demand volatilities on routes, as well as MGO or LNG price volatilities, we will conduct the same analysis to investigate the impacts of the policies under various market fluctuations.

4. Case study

Company SSS is a joint venture in Shanghai. Its main business scope covers the tanker shipping area. In this section, we use our theoretical models to analyze the fleet replacement decisions for its chemical tankers.

4.1. Parameters

4.1.1. Parameters of routes and demand

SSS currently has chemical tanker shipping business on three routes: Shanghai – Busan, Shanghai – Singapore and Shanghai – Brisbane. We look up the related website (e.g., http://app.cnss.com.cn/mileage_inquiry.php) to obtain the distances of these routes, which are shown in Table 1. Moreover, the historical shipping volume on these three routes is also shown in Table 1. Due to the sample limitation, we cannot use statistical methods to obtain the demand parameters on these routes. From the sample data in Table 1, we use the average volume increase rate on each route as the mean-drift of the demand, i.e., α_r . Moreover, we assume that their volatilities are 0.01 because of their relatively lower fluctuation.⁷ In the sensitivity analysis, we will show the impacts of the volatility changes on the vessel replacement decisions.

4.1.2. Parameters of MGO and LNG prices

Here, we use the method proposed by Dixit and Pindyck (1994) to estimate the parameters of MGO and LNG prices, i.e., η_0 , η_G , $\bar{F}_0$, $\bar{F}_G$, σ_0 and σ_G . First, the sample data is used to estimate the parameters a_0 , b_0 , a_G and b_G in the following equations by OLS:

$$F_0^t - F_0^{t-1} = a_0 + b_0 F_0^{t-1} + \varepsilon_0^t \quad (18)$$

$$F_G^t - F_G^{t-1} = a_G + b_G F_G^{t-1} + \varepsilon_G^t \quad (19)$$

Next, we can obtain the corresponding parameters as follows:

$$\bar{F}_0 = a_0 / b_0 \quad (20)$$

$$\eta_0 = -\log(1 + b_0) \quad (21)$$

$$\sigma_0 = \sigma_{\varepsilon,0} \sqrt{\frac{\log(1 + b_0)}{(1 + b_0)^2 - 1}} \quad (22)$$

$$\bar{F}_G = a_G / b_G \quad (23)$$

⁷ Although the volatility for the GBM is very low in our case study, its impacts on the fleet operation cost savings under different plans may be significant. In Section 4.2, we will show the average annual cost savings under different fleet compositions in Fig. 1. If we use the coefficient of variation (CV = standard deviation/mean) to reflect their fluctuations, we find that the CVs under the three fleet compositions are 0.070, 0.441 and 0.223, respectively. Although these CVs depend on the sample simulation paths, they can still show that the small volatility in demand may cause large fluctuations in the annual cost savings, which are crucial to the fleet replacement decisions. This is because the impacts of the small volatility in demand may be magnified by the relatively larger volatilities of the MGO and LNG prices, as well as the large demand base (the historical average demands on the three routes are 741,000, 1,957,000 and 293,000, respectively). Therefore, it is still worth modeling the demand as a stochastic process.

Table 1

Shipping volume at routes from 2012 to 2014. Source: From SSS.

Route	Distance (miles)	2012 (thousand tons)	2013 (thousand tons)	2014 (thousand tons)	Average increasing rate
Shanghai – Busan	482	729	731	764	0.02
Shanghai – Singapore	2172	1959	1878	2035	0.02
Shanghai – Brisbane	4216	299	274	305	0.01

$$\eta_G = -\log(1 + b_G) \quad (24)$$

$$\sigma_G = \sigma_{e,G} \sqrt{\frac{\log(1 + b_G)}{(1 + b_G)^2 - 1}} \quad (25)$$

where $\sigma_{e,0}$ and $\sigma_{e,G}$ are the standard deviations of regression Eqs. (18) and (19),

For the MGO price sample, we used the weekly data in Singapore from January 12th, 1990 to January 23th, 2015, with 1307 sample data points. For LNG prices, we use the weekly data from January 6th, 2012 to January 23th, 2015, with 157 sample data points.⁸ Based on these sample data, we use Eviews 6.0 to obtain the estimated values of a_0 , b_0 , a_G and b_G , and calculate the following parameters: $\bar{F}_0 = 6.1178$, $\eta_0 = 0.0022$, $\sigma_0 = 0.032$, $\bar{F}_G = 1.253$, $\eta_G = 0.0283$, $\sigma_G = 0.0402$.

Note that the simulated MGO and LNG prices are weekly data, while the simulated route demand is yearly data. To maintain correspondence with the demand data, we average the weekly MGO and LNG prices as their yearly levels.

4.1.3. Vessel parameters

Currently, there are two types of vessels in SSS, 13,000-DWT diesel-fuelled and 19,000-DWT diesel-fuelled chemical tankers. There are 5 vessels of each. The planned replacement vessels are 19,000-DWT dual-fuelled chemical tankers. The operating speeds for all vessels are 14 knots. Each diesel-fuelled vessel consumes 23.7 tons of MGO daily at sea and 9.4 tons of MGO daily at port. Each dual-fuelled vessel consumes 20.88 tons of LNG daily at sea and 3.12 tons of MGO daily at port. The daily fixed costs are the same for all vessels: \$5000/day. The lay-up costs for each type of vessel are: \$2226/day (13,000-DWT diesel-fuelled vessel), \$3767/day (19,000-DWT diesel-fuelled vessel), and \$4712/day (19,000-DWT dual-fuelled vessel). A 13,000-DWT diesel-fuelled vessel spends 9 days in berth on all three routes. A 19,000-DWT diesel-fuelled vessel and a 19,000-DWT dual-fuelled vessel both spend 13 days in berth on all three routes. The prices for a 13,000-DWT diesel-fuelled vessel, a 19,000-DWT diesel-fuelled vessel and a 19,000-DWT dual-fuelled vessel are \$12.25 million, \$27.5 million and \$34.4 million, respectively.⁹

4.1.4. Replacement plans

SSS wants to replace its current vessels with the 19,000-DWT dual-fuelled vessels. For simplicity, a replacement is implemented for all vessels of the specific type. E.g., if SSS considers replacing its 13,000-DWT diesel-fuelled vessels, all 5 vessels are replaced by five 19,000-DWT dual-fuelled vessels at one time. Therefore, in our case, the fleet replacement can be arranged as the following plans: ① replace the 13,000-DWT diesel-fuelled vessels first, and then the 19,000-DWT diesel-fuelled vessels; ② replace the 19,000-DWT diesel-fuelled vessels first, and then the 13,000-DWT diesel-fuelled vessels; and ③ replace all vessels at once. SSS will consider the vessel replacement decision for the future 10 years. These three replacement plans have different orders. Note that, in Plan ① and ②, SSS faces two decisions, with $(9 + 8 + 7 + \dots + 1 = 45)$ sequences in each plan. In Plan ③, he only needs to make one decision.

We consider the two vessel replacements in different years in Plan ① and Plan ②. In Plan ③, the two vessel replacements happen in one year. If we ignore the above consideration and assume that the two vessel replacements can be implemented in one year or different years, the “all at once” case (Plan ③) can be treated as a special case for both Plan ① and Plan ②. Thus, we need to compare 91 sequences (45 in Plan ①, 45 in Plan ② and the “all at once” case). Note that the 45 sequences in Plan ① and the other 45 sequences in Plan ② do not include the special “all at once” case here. These 91 sequences and their vessel replacement arrangements are listed in Table 2.

4.1.5. Other parameters

We here use the risk-free interest rate to represent the discount rate because the risk-neutral probabilities allow discounting at the risk-free rate and the expected values are discounted at the risk-free rate. The average risk-free interest rate is computed as 3.5% based on the previous rates of return on 10-year Treasury bonds over the past ten years (from January 2005 to January 2015). All previous interest rates are obtained from the website of the Federal Reserve Bank of St. Louis in the USA (<https://research.stlouisfed.org/fred2/series/GS10>). Moreover, the vessel life is 20 years. Therefore, the annuity factor of 20 years with a 3.5% discount rate is 0.068.

⁸ We obtain the MGO prices from Clarkson SIN (<http://www.clarksons.net>), and LNG prices from a Chinese energy consultant company in Beijing. The LNG prices are the Henry-hub prices based on \$/MMBTU. In our application, we convert them to \$/ton, which is consistent with the MGO prices.

⁹ The diesel-fuelled vessel price data are obtained from Clarkson SIN (<http://www.clarksons.net>). We use the latest price data (August, 2015). The dual-fuelled vessel price is obtained from a Chinese shipyard.

Table 2

The vessel replacement arrangements in the sequences.

Plan	Sequence No.	Vessel replacement arrangement	
①	1	Year 1	Year 2–Year 10
		Replacing 13,000DWT DE	Replacing 19,000DWT DE
	2	Year 1–Year 2	Year 3–Year 10
		Replacing 13,000DWT DE	Replacing 19,000DWT DE
	...	...	...
	10	Year 2	Year 3–Year 10
		Replacing 13,000DWT DE	Replacing 19,000DWT DE
	11	Year 2–Year 3	Year 4–Year 10
		Replacing 13,000DWT DE	Replacing 19,000DWT DE
	...	...	...
	②	46	Year 1
		Replacing 19,000DWT DE	Year 2–Year 10
	...	...	Replacing 13,000DWT DE
	...	...	...
	③	91	Year 1–Year 10
		Replacing all at once	

Notes: DE means diesel fuelled vessels.

In the application, we use 6th-order Hermite polynomials. The number of simulation paths in the multi-option LSM is 100. To reduce the bias, we make 10 runs for each parameter set and select their mean level as the result. The targeted tolerance is 5%, which equals the mean option value divided by its standard deviation.

The parameters used in the application are summarized in Table 3.

4.2. Replacement timing decisions

Using the above parameters, we obtain the option values of different plans within the future 10 years in Table 4. From the results, we find that Plan ③ has the highest option value. Moreover, we show the replacement probabilities in Plan ③ for each year in Table 4. From the results, SSS can obtain various management insights as follows:

- (i) It is better to replace all of his diesel-fuelled vessels with dual-fuelled vessels at one time, rather than replace them step-by-step. The reason is that the dual-fuelled vessels have much higher annual cost savings than the 13,000-DWT and 19,000-DWT diesel-fuelled vessels. Fig. 1 shows the average cost savings of the three fleet compositions on the 100 sample paths over 10 years compared to SSS's fleet (13,000-DWT diesel-fuelled vessels + 19,000-DWT diesel-fuelled vessels) now. In Plan ①, the fleet composition after the first replacement decision is 19,000-DWT diesel-fuelled vessels + 19,000-DWT dual-fuelled vessels. In Plan ②, the fleet composition after the first replacement decision is 13,000-DWT diesel-fuelled vessels + 19,000-DWT dual-fuelled vessels. In Plan ③, all vessels are 19,000-DWT dual-fuelled vessels after the replacement. From Fig. 1, we know that Plan ③ can yield higher average annual cost savings. Therefore, replacing the different diesel vessels at one time is better than replacing them step-by-step.
- (ii) There is a high possibility of SSS replacing his vessels within 5 years. The cumulative probability in the first 5 years of performing the fleet replacement is 83.8%. However, there is very little possibility of replacing the vessels in the first 3 years (the cumulative probability in the first 3 years is 11.8%). The better opportunities to perform the fleet replacement seem to be in the 4th or 5th year (see Table 5). We present our explanations from an example in Fig. 2, which shows the increasing rates of the average annual fleet operation cost savings of Plan ③ in one simulation run. The main idea for the real option approach is to compare the values of immediate replacement and deferring, or the value of replacing vessels later (see Eq. (10)). From Fig. 2, one can find that the 6th year is a turning point, where the increasing rate changes from positive to negative. This means that SSS can obtain more cost savings if he makes the fleet replacement in the 5th year than in the 6th year. One may argue that there is another turning point in the 8th year (from the negative to the positive) in Fig. 2. However, the real options theory tells us that decision-makers should exercise their options when a certain threshold is reached for the first time (Dixit and Pindyck, 1994). Therefore, SSS should not wait until the second opportunity (the 8th year) to make the fleet replacement (exercise the option). Although Fig. 2 only shows the results in one simulation run, which is not enough to represent the entire picture, it can help us to understand the logic of the results.

4.3. Sensitivity analysis and comparisons of policy performances

To investigate the impact of the changing parameters (e.g., the demand volatilities on routes and the fuel price volatilities) on SSS's fleet replacement decisions and compare the performances of government policies (the subsidy for building new energy vessels versus the subsidy for using LNG as fuel) under different scenarios, we calculate the replacement probabilities

Table 3

Parameters values in the application.

Parameter	Concept	Value
r	Routes	Shanghai – Busan, Shanghai – Singapore, Shanghai – Brisbane
v	Vessel types	13,000 DWT diesel fuelled vessel, 19,000 DWT diesel fuelled vessel,
K_v	Vessel capacity	19,000 DWT dual fuelled vessel
α_r	Mean-drift of route demand	Shanghai – Busan: 0.02 Shanghai – Singapore: 0.02 Shanghai – Brisbane: 0.01
σ_r	Volatility of route demand	Shanghai – Busan: 0.01 Shanghai – Singapore: 0.01 Shanghai – Brisbane: 0.01
$\bar{F}_O$	Mean level of the MGO price	6.1178
η_O	Reversion speed parameter of MGO price	0.0022
σ_O	Volatility parameter of MGO price	0.032
$\bar{F}_G$	Mean level of the LNG price	1.253
η_G	Reversion speed parameter of LNG price	0.0283
σ_G	Volatility parameter of LNG price	0.0402
Dis_r	Route voyage distance	Shanghai – Busan: 482 miles Shanghai – Singapore: 2172 miles Shanghai – Brisbane: 4216 miles
Sp_v	Ship sailing speed	13,000 DWT diesel fuelled vessel: 14 knots 19,000 DWT diesel fuelled vessel: 14 knots 19,000 DWT dual fuelled vessel: 14 knots
C_v^L	Ship lay-up costs	13,000 DWT diesel fuelled vessel: \$2226/day 19,000 DWT diesel fuelled vessel: \$3767/day 19,000 DWT dual fuelled vessel: \$4712/day
OCS_v	MGO consumption rate of the ship when sailing	13,000 DWT diesel fuelled vessel: 23.7 tons/day 19,000 DWT diesel fuelled vessel: 23.7 tons/day
OCB_v	MGO consumption rate of the ship in berth	13,000 DWT diesel fuelled vessel: 9.4 tons/day 19,000 DWT diesel fuelled vessel: 9.4 tons/day 19,000 DWT dual fuelled vessel: 3.12 tons/day
GCS_v	LNG consumption rate of the ship when sailing	19,000 DWT dual fuelled vessel: 20.88 tons/day
BD_{vr}	Days in berth for different ships in different routes	13,000 DWT diesel fuelled vessel Shanghai – Busan: 9 days Shanghai – Singapore: 9 days Shanghai – Brisbane: 9 days 19,000 DWT diesel fuelled vessel Shanghai – Busan: 13 days Shanghai – Singapore: 13 days Shanghai – Brisbane: 13 days 19,000 DWT Dual fuelled vessel Shanghai – Busan: 13 days Shanghai – Singapore: 13 days Shanghai – Brisbane: 13 days
DFC_v	Daily fixed cost of the ship	\$5000/day for all types of vessels
N_v	Number of ships available	13,000 DWT diesel fuelled vessel: 5 19,000 DWT diesel fuelled vessel: 5 19,000 DWT dual fuelled vessel: 5
N_v^I	Number of ships available in the market for chartering in	13,000 DWT diesel fuelled vessel: 10 19,000 DWT diesel fuelled vessel: 10
C_v^I, C_v^O	Costs of ship chartering in and out	13,000 DWT diesel fuelled vessel: \$9750/day 19,000 DWT diesel fuelled vessel: \$15,500/day 19,000 DWT dual fuelled vessel: \$19,500/day
F_r	Route frequency requirement	52 for every route
ρ	Discount rate	3.5%
AF	Annuity factor	0.068
P_{MGO}	Diesel fuelled vessel price	13,000 DWT: \$12.25 million 19,000 DWT: \$27.5 million
P_{LNG}	Dual fuelled vessel price	\$34.4 million
N	Order of the Hermite polynomials	6
Ω	Number of the simulation paths	100

Table 4

Option values of different plans.

Plan	Mean	S.D	CV
④	9,594,913	342,538	0.0357
⑤	6,658,702	320,949	0.0482
⑥	10,872,469	540,051	0.0497

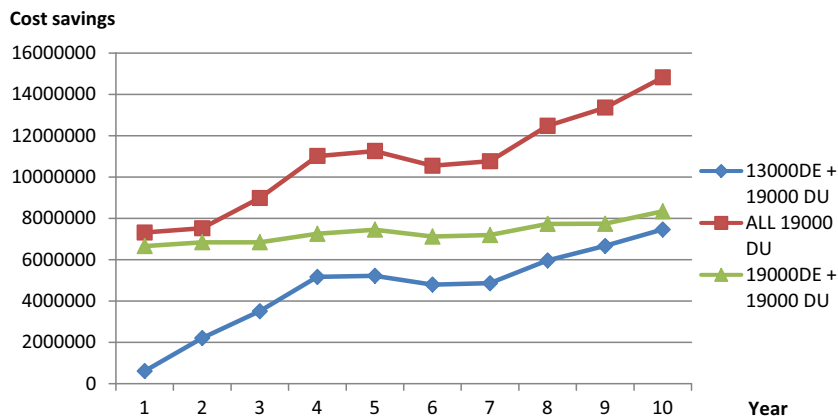


Fig. 1. Average cost savings of the different fleet compositions compared to the original fleet (13,000DWT DE + 19,000DWT DE). Notes: DE: diesel fuelled vessels; DU: dual fuelled vessels.

Table 5

Replacement probabilities in Plan ③ for each year.

Year	1	2	3	4	5	6	7	8	9	10
Probability	0.007	0.057	0.054	0.450	0.270	0.047	0.011	0.004	0.014	0.084

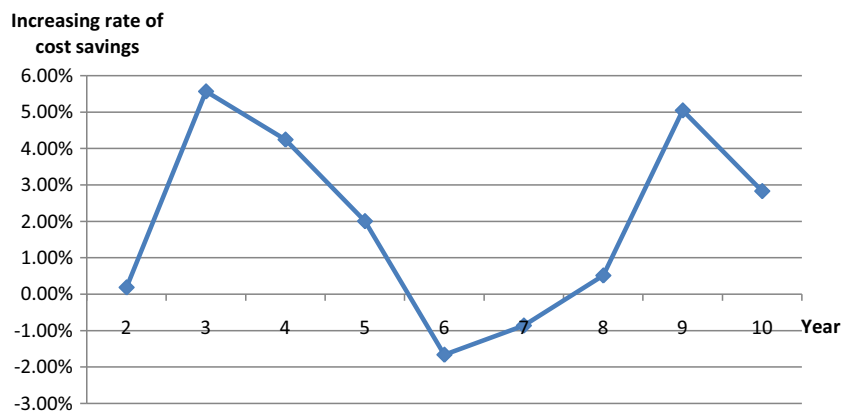


Fig. 2. Increasing rates of the average annual fleet operation cost savings of Plans ③ in one simulation run.

and the expanded NPVs of the cost savings with the change of σ_O , σ_G and σ_r . The results¹⁰ are summarized in Table 6 and Figs. 3–6. From the results, we can obtain various policy implications as follows:

- (i) The vessel subsidy policy has larger impacts on the expected NPVs of the cost savings and the fleet replacement timing than the fuel subsidy policy. From Table 6 and Figs. 3–6, we know that the vessel subsidy policy can yield increased cost savings and earlier replacement decisions than the fuel subsidy policy in nearly all scenarios given the same price reduction amplitude. This means that the shipowner prefers the vessel subsidy policy because it has an immediate reward to him. Although the fuel subsidy policy can also increase the shipowner's cost savings, its impacts are little-by-little and depend on the uncertain future (the future LNG prices). Therefore, the shipowner prefers to obtain the present benefit rather than waiting for a risky future reward.
- (ii) These two policies have different robustness performances. From Figs. 3–6, we know that the vessel subsidy policy has relatively stable impacts on the shipowner's replacement decisions in various scenarios, while the fuel subsidy policy has more volatile performance, especially when the demand volatility in the Shanghai – Singapore route changes (see Fig. 6). When the demand volatility in this route changes from +40% to –40%, the fleet replacement probability in the

¹⁰ We omit the results under the change of the demand volatility on the route of Shanghai to Brisbane because its impacts on SSS's fleet replacement decisions are very limited (the results are almost the same as with the benchmark case).

Table 6

Impacts to the expanded NPV by the vessel subsidy policy and the fuel subsidy policy in different scenarios.

Policy	Vessel subsidy policy		Fuel subsidy policy	
Scenario	Dual fuelled ship price –10%	Dual fuelled ship price –10%	LNG price –10%	LNG price –20%
<i>Change of the LNG price volatility</i>				
+10%	17.28%	35.69%	6.53%	9.98%
+20%	16.30%	35.19%	6.16%	9.30%
0%	17.40%	33.63%	7.50%	11.73%
–10%	19.22%	39.13%	7.69%	9.27%
–20%	18.65%	34.63%	7.42%	9.05%
<i>Change of the MGO price volatility</i>				
+10%	18.33%	34.42%	6.76%	11.57%
+20%	18.83%	34.84%	4.91%	11.67%
0%	17.40%	33.63%	7.50%	11.73%
–10%	16.60%	33.54%	3.50%	8.15%
–20%	17.28%	37.06%	6.90%	10.48%
<i>Change of the demand volatility in Shanghai – Busan route</i>				
+10%	14.58%	33.43%	3.95%	9.94%
+20%	17.71%	30.41%	4.30%	10.23%
0%	17.40%	33.63%	7.50%	11.73%
–10%	16.19%	35.02%	7.02%	11.62%
–20%	17.54%	35.27%	6.08%	11.04%
<i>Change of the demand volatility in Shanghai – Singapore route</i>				
+10%	15.31%	32.58%	3.21%	4.20%
+20%	14.28%	30.10%	6.07%	8.25%
0%	17.40%	33.63%	7.50%	11.73%
–10%	15.74%	34.99%	4.11%	9.90%
–20%	17.17%	36.45%	7.59%	12.19%

Note: The impacts of the change of the demand volatility in Shanghai – Brisbane route are very limited. Therefore, we omit these results in the table.

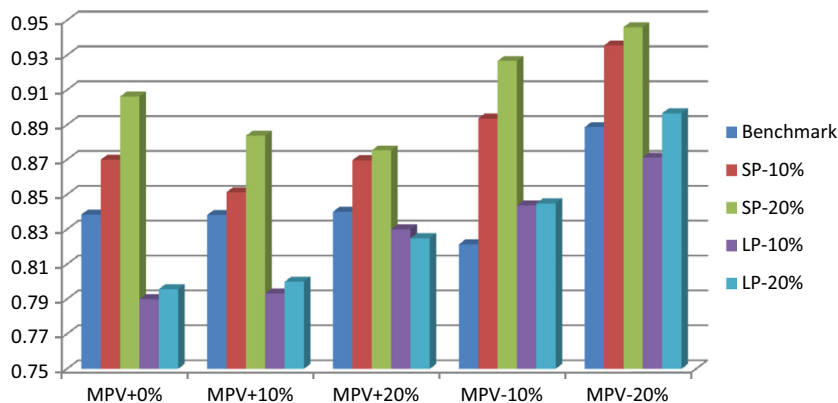


Fig. 3. Impacts to the fleet replacement probability in the first 5 years by the vessel subsidy policy and the fuel subsidy policy as the MGO price volatility changes. Notes: MPV: MGO price volatility; SP: Ship price; LP: LNG price.

first 5 years changes from 80.4% to 96.0% under the 20% vessel price subsidy policy, whereas it changes from 69.3% to 95.8% under the 20% fuel price subsidy policy. This is because the vessel subsidy is related to the relatively stable vessel prices, while the fuel subsidy could vary severely with the more volatile LNG prices. Moreover, SSS has the largest turnover volume on the Shanghai – Singapore route,¹¹ which means it uses more fuel on this route than the other routes. Therefore, the fuel subsidy policy is suitable on the routes with larger turnover volume, if governments wish to see the more obvious policy feedbacks.

¹¹ We use the year 2014 as an example to show this point. The Shanghai – Busan route has 368,118 ton miles, the Shanghai – Singapore route has 4,420,063 ton miles, and the Shanghai – Brisbane route has 1,286,723 ton miles (see Table 1).

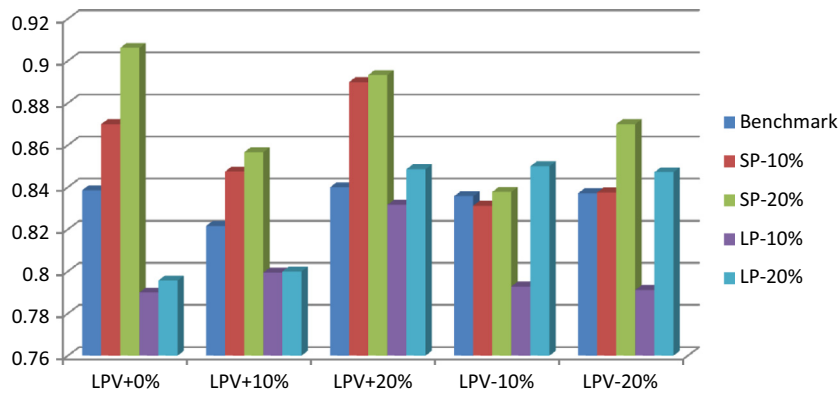


Fig. 4. Impact to the fleet replacement probability in the first 5 years by the vessel subsidy policy and the fuel subsidy policy as the LNG price volatility changes. Notes: LPV: LNG price volatility; SP: Ship price; LP: LNG price.

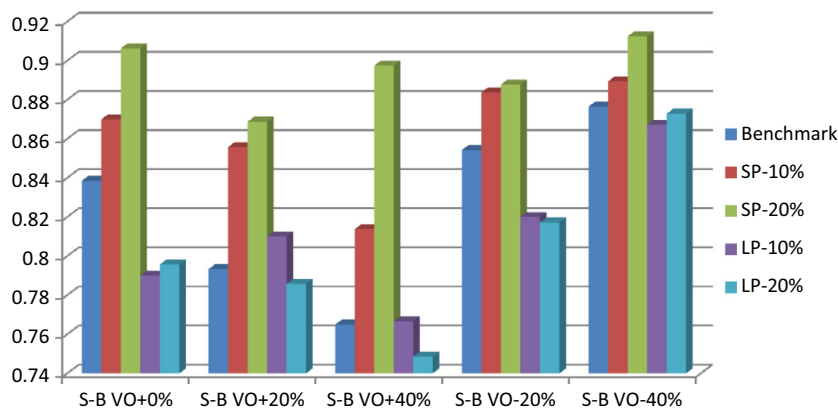


Fig. 5. Impact to the fleet replacement probability in the first 5 years by the vessel subsidy policy and the fuel subsidy policy as the demand volatility in the Shanghai – Busan route changes. Notes: S-B VO: the demand volatility in the Shanghai – Busan route; SP: Ship price; LP: LNG price.

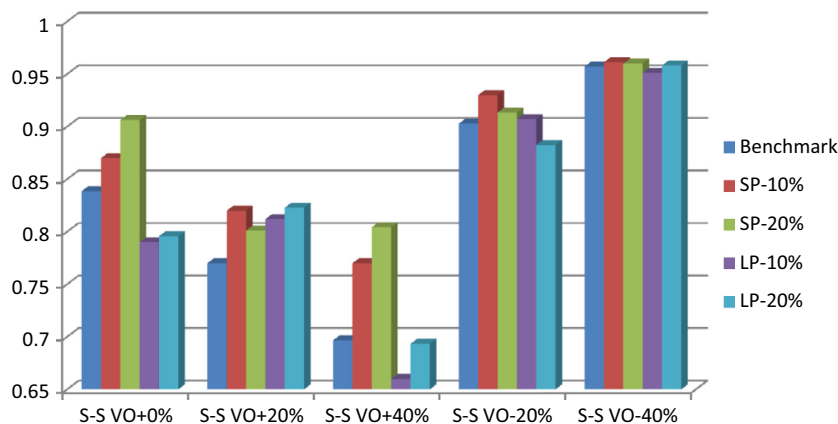


Fig. 6. Impact to the fleet replacement probability in the first 5 years by the vessel subsidy policy and the fuel subsidy policy as the demand volatility in the Shanghai – Singapore route changes. Notes: S-S VO: the demand volatility in the Shanghai – Singapore route; SP: Ship price; LP: LNG price.

5. Conclusions and future research

In this paper, we present a real option model to analyze the fleet replacement timing decisions when a shipowner faces uncertain demand on various routes and volatile fuel prices. Moreover, a multi-option least-squares Monte Carlo simulation

algorithm is used to find the replacement probabilities in future years and the expected cost savings made by the replacement decision. The proposed model is applied to a chemical tanker shipping company. The results reveal that our model is easy to implement and can provide support for shipowners in performing fleet replacement. In our case study, it is found that the shipowner should replace all of his diesel-fuelled vessels with dual-fuelled vessels at one time, rather than replace them step-by-step. The future 4th and 5th years are the most beneficial years to perform the fleet replacement. Moreover, we compare the performance of two possible government policies, the vessel subsidy policy and the fuel subsidy policy. The sensitivity analysis shows that, in our case, the shipowner prefers the vessel subsidy policy, which provides him increased cost savings and enables him to make earlier replacement decisions. Alternately, the fuel subsidy policy is suitable on the routes with larger turnover volume, if governments wish to see the more obvious policy feedbacks.

Some possible extensions could be considered in future research. The fleet deployment model can be extended to consider more issues (e.g., restrictions on calling port orders and time windows) and their impacts on fleet replacement outcomes. In addition, we only consider a small-scale case with three routes and three types of vessels. If the number of routes (or even for a complicated hub-spoke network) and vessel types were to increase, new algorithm designs should be investigated. Technically, our work is based heavily on Chow and Regan's methodology. It is worth noting that some of its shortcomings. The method is computationally costly at higher numbers of projects. [Chow and Sayarshad \(forthcoming\)](#) sample the maximum value to avoid explicit enumeration of all sequences. Also, [Chow et al. \(2011\)](#) introduced a greedy heuristic for improved computational performance, and also discuss a counter force to deferral, which is the first mover advantage if the decision is considered jointly with another player in the market. These improvements can be integrated into the algorithm design of the fleet replacement problem under uncertainty in the future work.

Acknowledgements

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Appendix A. An illustrated example of one vessel replacement under uncertainty

In [Appendix A](#), we use a small example with one vessel to show the implementation of our models. In such an example, we first solve it analytically. Then, we generate 10 sample paths and use the LSM algorithm to solve it again. Finally, the results from the two approaches are compared to show the correctness of our models and method.

A.1. Problem description

Here, we consider a replacement decision for one vessel on one route. The capacity of the diesel-fuelled vessel is 10,000 DWT. The dual-fuelled vessel has the same capacity and speed as the diesel-fuelled vessel. The time spent on a single voyage is 10 days. The annual savings of the operational costs after the vessel replacement can be expressed as follows:

$$V = D \cdot \Delta C \quad (26)$$

where D is the annual demand on the route and ΔC is the annual savings of the operational costs for every DWT after vessel replacement.

$$\Delta C = \frac{VT \cdot (OCS \cdot F_O - GCS \cdot F_G)}{K} \quad (27)$$

where VT is the voyage time, K is the vessel capacity. OCS_v and GCS_v are the MGO and LNG consumption rates of the diesel-fuelled and dual-fuelled vessels, respectively, and F_O and F_G are the MGO and LNG prices, respectively. For simplicity, we assume that the dual-fuelled ship consumes only LNG on its voyage. Both vessels have the same fuel consumption rates on their sailing and berth days. In this example, we assume that $OCS = GCS = 20$ tons/day, $F_O = \$250$ /ton and $F_G = \$200$ /ton.

Moreover, we suppose D follows a GBM as follows:

$$dD = \alpha D dt + \sigma D dW \quad (28)$$

where $\alpha = 0.01$ and $\sigma = 0.5$.

The annual cost increase for the vessel replacement is $\Delta P = \$350,000$, which is the annuity of the price difference between a dual-fuelled ship and a diesel-fuelled ship.

Thus, the shipowner faces the following optimization problem:

$$F = \max_T [(V_T - \Delta P)e^{-\rho T}] \quad (29)$$

where F is the option value of replacement, T is the vessel replacement time, V_T is the annual cost savings after the vessel replacement at time T , and $\rho = 0.04$ is the discount rate.

A.2. Closed-form solution

Using the approach proposed by Dixit and Pindyck (1994), we obtain the optimal vessel replacement rule as follows:

$$V^* = \frac{\beta_1}{\beta_1 - 1} \Delta P \quad (30)$$

where β_1 is the solutions of the following quadratic equation and $\beta_1 > 1$.

$$1/2\sigma^2\beta(\beta - 1) + \alpha\beta - \rho = 0. \quad (31)$$

The replacement rule means that the shipowner should replace his diesel-fuelled vessel only if the annual savings of the operational costs reach the threshold V^* . Moreover, the option value of vessel replacement is

$$F^* = V^* - \Delta P \quad (32)$$

Using the parameter values in Appendix A.1, we obtain the theoretical option value of vessel replacement $F^* = \$254,949$.

A.3. LSM simulation

Now, we generate 10 sample paths. The annual demands in the future ten years on these 10 sample paths are shown in Table A-1. Then, we obtain the annual cost savings $V_t - \Delta P$ after the vessel replacement in these paths, which are shown in Table A-2. In each year, the shipowner needs to decide whether to replace the vessel immediately or defer the decision to the next year. We use Year 4 as an example to show how to make the decision. The main idea is to compare the payoff from the immediate replacement (i.e., $V_t - \Delta P$) to the value of deferring replacement (i.e., the expected discounted $V_5 - \Delta P$) in the sample paths, where $V_4 - \Delta P > 0$. Longstaff and Schwartz (2001) call these paths with positive payoffs from immediate exercising the “in-the-money paths.” To calculate the expected discounted $V_5 - \Delta P$, we regress¹² the discounted $V_5 - \Delta P$ on a constant, $V_4 - \Delta P$, and $(V_4 - \Delta P)^2$. The corresponding data are shown in Table A-3. The regression equation is expressed as follows.

$$Y = -316,840 + 5.69X - 1.11 \times 10^{-5}X^2 \quad (33)$$

where Y is the discounted $V_5 - \Delta P$ and X is $V_4 - \Delta P$.

Substituting $V_4 - \Delta P$ into the in-the-money paths into (35), we obtain the expected discounted $V_5 - \Delta P$ (Column 4 in Table A-3). Comparing $V_4 - \Delta P$ (Column 2 in Table A-3) to the expected discounted $V_5 - \Delta P$, we obtain the optimal early replacement decision at Year 4, where 1 means replacement and 0 means deferring the decision (Column 5 in Table A-3).

We begin the recursion proposed by the LSM algorithm from Year 10. At year 10, the values of deferring replacement in all paths are 0. We repeat the above process recursively and obtain the optimal early replacement decisions for all years, which are shown in the intermediate decision matrix in Table A-4. By examining the intermediate decision matrix, the final vessel replacement decision is made until the first decision year occurs along each sample path, which is shown in Table A-5. According to Table A-5, we can obtain the vessel replacement probabilities for each year (Table A-6). The option value is valued by discounting the annual cost savings from the decision year to Year zero and then taking the average over all 10 paths: $F^{LSM} = \$278,300$, where the superscript LSM means the simulated option value.¹³ We perform 10 simulation runs for the above parameter set and select their mean level as the result. Meanwhile, we further use 30 and 100 paths to repeat our simulations. The results under 10, 30 and 100 paths are summarized in Table A-7, where one can find that the simulation results converge to the theoretical result F^* as the number of paths increases.

Appendix B. An illustrated example of two different vessel replacements under uncertainty

In Appendix B, we illustrate a small example with two different vessel replacements under uncertainty. There is no analytical approach for multi-option cases. Therefore, we only show the implementation of our models.

B.1. Problem description

We now consider a replacement decision for a fleet with two different vessels on one route. The capacities of the diesel-fuelled vessels are 4000 DWT (Vessel A) and 6000 DWT (Vessel B). These two vessels are operated on one route at the same speed. The time spent on a single voyage is 10 days for both vessels. There are two dual-fuelled vessels that have the same capacities and speeds as the corresponding diesel-fuelled vessels. We assume that the replacements only occur with vessels

¹² Because of the limited sample (5 in-the-money paths), we use the binominal regression equation, as used in the numerical example in Longstaff and Schwartz (2001), who demonstrate that this simple regression can also yield relatively accurate results.

¹³ It is worth noting that the analytical model (29) is under an infinite horizon. Therefore, the simulated option value with 10 years should technically be lower than the theoretical result since there is less flexibility in the case of 10 years. Here, $F^{LSM} = \$278,300 > F^* = \$254,949$ because it is the result of one simulation run. In Table A-7, it is shown that the mean levels of the 10 simulation runs are all less than the theoretical result F^* under different numbers of paths. We thank one of the anonymous reviewers for pointing this out for us.

Table A-1

The simulated annual demand in the 10 paths.

Path	Year									
	1	2	3	4	5	6	7	8	9	10
1	395,782	591,655	837,056	770,584	418,586	177,780	436,030	537,217	327,833	303,844
2	632,101	651,863	664,007	799,063	396,202	345,507	94,656	68,199	45,854	79,253
3	315,525	190,640	234,210	105,685	39,755	57,292	39,574	59,589	70,108	56,174
4	304,600	360,448	397,458	421,528	433,980	922,532	424,669	526,717	355,411	296,200
5	730,406	326,128	150,645	122,610	49,250	35,404	22,960	71,746	40,877	20,285
6	241,792	243,548	176,286	98,233	92,522	36,554	38,195	57,942	42,089	18,768
7	311,533	416,017	684,076	598,426	790,634	765,868	731,453	1,162,693	956,280	995,560
8	494,164	489,991	427,347	147,591	131,411	141,382	88,331	80,849	88,434	69,582
9	336,244	465,297	554,955	170,699	159,406	126,847	166,747	78,045	43,206	49,543
10	629,798	1,555,944	838,478	414,937	306,142	153,637	186,940	138,404	144,581	82,476

Table A-2

The annual cost savings after the vessel replacement in the 10 simulation paths.

Path	Year									
	1	2	3	4	5	6	7	8	9	10
1	45,782	241,655	487,056	420,584	68,586	−172,220	86,030	187,217	−22,167	−46,156
2	282,101	301,863	314,007	449,063	46,202	−4493	−255,344	−281,801	−304,146	−270,747
3	−34,475	−159,360	−115,790	−244,315	−310,245	−292,708	−310,426	−290,411	−279,892	−293,826
4	−45,400	10,448	47,458	71,528	83,980	572,532	74,669	176,717	5411	−53,800
5	380,406	−23,872	−199,355	−227,390	−300,750	−314,596	−327,040	−278,254	−309,123	−329,715
6	−108,208	−106,452	−173,714	−251,767	−257,478	−313,446	−311,805	−292,058	−307,911	−331,232
7	−38,467	66,017	334,076	248,426	440,634	415,868	381,453	812,693	606,280	645,560
8	144,164	139,991	77,347	−202,409	−218,589	−208,618	−261,669	−269,151	−261,566	−280,418
9	−13,756	115,297	204,955	−179,301	−190,594	−223,153	−183,253	−271,955	−306,794	−300,457
10	279,798	1,205,944	488,478	64,937	−43,858	−196,363	−163,060	−211,596	−205,419	−267,524

Table A-3

The optimal early replacement decision at Year 4.

Path	$V - \Delta P$ at Year 4 in the in-the-money paths	Discounted $V - \Delta P$ at Year 5 in the in-the-money paths	Expected $V - \Delta P$ at Year 4 if not replacing the vessel	Replacement decision
1	420,584	65,948	116,216	1
2	449,063	44,425	3776	1
3	−	−	−	0
4	71,528	80,750	33,572	1
5	−	−	−	0
6	−	−	−	0
7	248,426	423,687	413,046	0
8	−	−	−	0
9	−	−	−	0
10	64,937	−42,172	6028	1

Table A-4

The intermediate decision matrix.

Path	Year									
	1	2	3	4	5	6	7	8	9	10
1	0	0	1	1	0	0	0	1	0	0
2	0	0	1	1	1	0	0	0	0	0
3	0	0	0	0	0	0	0	0	0	0
4	0	0	1	1	0	1	0	1	1	0
5	1	0	0	0	0	0	0	0	0	0
6	0	0	0	0	0	0	0	0	0	0
7	0	0	1	0	1	1	0	1	0	1
8	0	0	1	0	0	0	0	0	0	0
9	0	0	1	0	0	0	0	0	0	0
10	0	1	1	1	0	0	0	0	0	0

of the same capacity, i.e., the 4000-DWT diesel-fuelled vessel can only be replaced by a 4000-DWT dual-fuelled vessel, if necessary. The fuel consumption rates of the two diesel-fuelled vessels and the two dual-fuelled vessels are the same as in

Table A-5

The vessel replacement decisions in the 10 simulation paths for each year.

Path	Year									
	1	2	3	4	5	6	7	8	9	10
1	0	0	1	0	0	0	0	0	0	0
2	0	0	1	0	0	0	0	0	0	0
3	0	0	0	0	0	0	0	0	0	0
4	0	0	1	0	0	0	0	0	0	0
5	1	0	0	0	0	0	0	0	0	0
6	0	0	0	0	0	0	0	0	0	0
7	0	0	1	0	0	0	0	0	0	0
8	0	0	1	0	0	0	0	0	0	0
9	0	0	1	0	0	0	0	0	0	0
10	0	1	0	0	0	0	0	0	0	0

Table A-6

The vessel replacement probabilities for each year.

Year	1	2	3	4	5	6	7	8	9	10
Probability	0.1	0.1	0.6	0	0	0	0	0	0	0

Table A-7

Comparison of the option values under different numbers of paths.

Path	Mean	S.D	CV
10	198,507	56,814	0.29
30	209,896	48,766	0.23
100	216,624	25,972	0.12

[Appendix A](#), i.e., $OCS = GCS = 20$ tons/day for all vessels on their sailing and berth days. The fuel prices are also the same as in [Appendix A](#). Using (27), we can calculate the annual savings of the operational costs for every DWT after the vessel replacement for Vessel A and Vessel B as follows: $\Delta C_A = \$2.5$ and $\Delta C_B = \$1.67$. Moreover, we assume that the market shares for these two vessels depend on their capacities, i.e., 40% for Vessel A (and its replacement) and 60% for Vessel B (and its replacement). We use the same equation (28) and parameters as in [Appendix A](#) to describe the route demand D . The annual cost increases for the vessel replacements are $\Delta P_A = \$160,000$ and $\Delta P_B = \$200,000$, respectively. Here, the discount rate is the same as in [Appendix A](#).

B.2. Multi-option LSM simulation

We consider the following replacement plans: ① replace the 4000-DWT diesel-fuelled vessels first, and then the 6000-DWT diesel-fuelled vessels in one of the later years; ② replace the 6000-DWT diesel-fuelled vessels first, and then the 4000-DWT diesel-fuelled vessels in one of the later years; ③ replace all at once in one year. We consider the vessel replacement decisions in the future 10 years. These three replacement plans have different orders. In Plan ① and ②, we face two decisions, with 45 sequences in each plan. In Plan ③, we only need to make one decision. Therefore, we should compare the option values for 91 sequences and find the maximum one. The sequences here are similar to those in [Section 4.1.4](#).

Now, we generate 10 sample paths. Using (26), we can obtain the annual cost savings for the fleet after the vessel replacements in these paths. [Table B-1–B-3](#) show the results after replacing Vessel A, Vessel B and Vessel A and B simultaneously in different years, respectively.

We use Sequence 3 as an example to show how to make the decision. In Sequence 3, we should replace Vessel A at one year between Year 1 and Year 3 and Vessel B at one year between Year 4 and Year 10. First, we analyze the decision to replace Vessel B. The analysis procedure is similar to that in [Appendix A](#) because there are no other decisions after it. For instance, at Year 4, we choose the “in-the-money paths” from Column 5 in [Table B-3](#) and copy them to Column 2 in [Table B-4](#). The discounted values of the next year (Column 6 in [Table B-3](#)) in the in-the-money paths are shown in Column 3 of [Table B-4](#). The regression equation is expressed as follows:

$$Y = 291,777 + 0.61X - 9.6 \times 10^{-9}X^2 \quad (34)$$

where Y is the values of Column 3 in [Table B-4](#) and X is the values of Column 2 in [Table B-4](#). Using (34), we obtain the expected discounted values for delaying replacement, which are shown in Column 4 of [Table B-4](#). Comparing the values of immediate replacement (Column 2 of [Table B-4](#)) and the values of delaying replacement (Column 4 of [Table B-4](#)), we

Table B-1

The annual cost savings to the fleet after replacing Vessel A in the 10 simulation paths.

Path	Year									
	1	2	3	4	5	6	7	8	9	10
1	197,008	412,387	238,741	302,471	458,778	374,510	185,847	95,816	89,120	3122
2	239,445	219,385	148,950	81,787	123,107	−68,179	42,204	110,974	47,114	51,810
3	−11,383	−64,611	−73,002	−55,315	−104,845	−129,914	−144,371	−139,228	−142,666	−139,132
4	27,317	−80,393	−91,268	−77,551	−70,350	−51,455	−53,114	−58,824	−38,545	−46,626
5	89,494	80,370	130,862	−73,093	−106,809	−113,871	−140,780	−137,205	−125,706	−87,405
6	169,561	282,322	256,421	31,172	203,693	25,294	−44,978	−33,969	−58,247	−93,095
7	388,563	262,423	772,636	244,442	194,683	71,143	−6851	−76,660	−72,489	−106,040
8	109,873	23,563	31,248	48,407	260,941	265,069	303,089	443,474	317,495	17,437
9	426,228	287,818	824,008	1,669,385	1,158,442	555,296	861,230	495,345	495,213	760,762
10	235,696	43,843	−33,431	−64,189	−45,325	6430	12,400	−46,380	−33,802	13,538

Table B-2

The annual cost savings to the fleet after replacing Vessel B in the 10 simulation paths.

Path	Year									
	1	2	3	4	5	6	7	8	9	10
1	157,008	372,387	198,741	262,471	418,778	334,510	145,847	55,816	49,120	−36,878
2	199,445	179,385	108,950	41,787	83,107	−108,179	2204	70,974	7114	11,810
3	−51,383	−104,611	−113,002	−95,315	−144,845	−169,914	−184,371	−179,228	−182,666	−179,132
4	−12,683	−120,393	−131,268	−117,551	−110,350	−91,455	−93,114	−98,824	−78,545	−86,626
5	49,494	40,370	90,862	−113,093	−146,809	−153,871	−180,780	−177,205	−165,706	−127,405
6	129,561	242,322	216,421	−8828	163,693	−14,706	−84,978	−73,969	−98,247	−133,095
7	348,563	222,423	732,636	204,442	154,683	31,143	−46,851	−116,660	−112,489	−146,040
8	69,873	−16,437	−8752	8407	220,941	225,069	263,089	403,474	277,495	−22,563
9	386,228	247,818	784,008	1,629,385	1,118,442	515,296	821,230	455,345	455,213	720,762
10	195,696	3843	−73,431	−104,189	−85,325	−33,570	−27,600	−86,380	−73,802	−26,462

Table B-3

The annual cost savings to the fleet after replacing Vessel A and Vessel B simultaneously in the 10 simulation paths.

Path	Year									
	1	2	3	4	5	6	7	8	9	10
1	354,015	784,775	437,481	564,943	877,555	709,020	331,693	151,633	138,239	−33,755
2	438,890	398,769	257,901	123,573	206,214	−176,357	44,408	181,947	54,228	63,619
3	−62,766	−169,222	−186,004	−150,630	−249,691	−299,827	−328,742	−318,457	−325,332	−318,264
4	14,633	−200,786	−222,537	−195,103	−180,701	−142,910	−146,229	−157,648	−117,091	−133,251
5	138,988	120,740	221,723	−186,186	−253,618	−267,742	−321,561	−314,409	−291,412	−214,810
6	299,123	524,644	472,842	22,345	367,385	10,588	−129,955	−107,939	−156,494	−226,191
7	737,125	484,846	1,505,272	448,884	349,366	102,286	−53,702	−193,320	−184,977	−252,080
8	179,746	7126	22,496	56,814	481,882	490,138	566,178	846,949	594,989	−5127
9	812,456	535,637	1,608,017	3,298,770	2,276,884	1,070,592	1,682,460	950,691	950,427	1,481,525
10	431,391	47,686	−106,862	−168,377	−130,651	−27,141	−15,201	−132,761	−107,603	−12,924

obtain the replacement decisions at Year 4, which are shown in Column 5 of Table B-4. Using the same procedure, we obtain the replacement decisions for Vessel B with Year 4 to Year 10 in all sample paths. The results are shown from Column 5 to Column 11 of Table B-6.

Next, we analyze the decision to replace Vessel A. We use Year 1 as an example. We choose the “in-the-money paths” from Column 2 of Table B-1 and copy them to Column 2 of Table B-5. If we replace Vessel A (exercise the first option) in these paths, we have the opportunity to replace Vessel B (exercise the subsequent option) between Year 4 and Year 10. From Column 5 to Column 11 of Table B-6, we know the replacement decisions for Vessel B in the sample paths. Therefore, we can calculate the discounted annual cost savings of replacing Vessel B at Year 1 in these in-the-money paths, which are shown in Column 5 of Table B-5. We regress the payoff of replacing Vessel A at Year 1 and the discounted payoff of replacing Vessel B at year 1. The equation is expressed as follows:

$$Y = 410,705 - 5.28X + 2.08 \times 10^{-5}X^2 \quad (35)$$

where Y is the values of Column 5 in Table B-5 and X is the values of Column 2 in Table B-5. Using (35), we obtain the expected discounted values for replacing Vessel B, which are shown in Column 6 of Table B-5. Meanwhile, we calculate the discounted values of Year 2 (Column 3 of Table B-1) in the in-the-money paths, which are shown in Column 3 of

Table B-4

The optimal early replacement decision at Year 4 for Sequence 3.

Path	The annual cost savings at Year 4 in the in-the-money paths	Discounted annual cost savings at Year 5 in the in-the-money paths	Expected annual cost savings at Year 4 if not replacing the vessel	Replacement decision
1	564,943	843,803	631,926	0
2	123,573	198,283	366,706	0
3	–	–	–	–
4	–	–	–	–
5	–	–	–	–
6	22,345	353,255	305,349	0
7	448,884	335,929	562,549	0
8	56,814	463,348	326,264	0
9	3,298,770	2,189,311	2,191,136	1
10	–	–	–	–

Table B-5

The optimal early replacement decision at Year 1 for Sequence 3.

Path	The annual cost savings at Year 1 in the in-the-money paths	Discounted annual cost savings at Year 2 in the in-the-money paths	Expected annual cost savings at Year 1 if delaying the replacement	Discounted annual cost savings of replacing Vessel B in the in-the-money paths	Expected annual cost savings of replacing Vessel B at Year 1	Replacement decision
1	197,008	396,526	207,051	750,138	179,687	1
2	239,445	210,947	242,372	176,273	341,698	1
3	–	–	–	–	–	0
4	27,317	–77,301	–61,316	0	282,066	1
5	89,494	77,279	60,628	0	105,232	1
6	169,561	271,463	177,433	314,042	114,881	1
7	388,563	252,330	265,601	298,640	1,506,280	1
8	109,873	22,657	94,653	411,915	82,338	1
9	426,228	276,748	246,621	2,932,595	1,947,060	1
10	235,696	42,156	239,763	0	324,361	1

Table B-6

The vessel replacement decisions in the 10 simulation paths at each year for Sequence 3.

Path	Year									
	1	2	3	4	5	6	7	8	9	10
1	1	0	0	0	2	0	0	0	0	0
2	1	0	0	0	2	0	0	0	0	0
3	0	0	0	0	0	0	0	0	0	0
4	1	0	0	0	0	0	0	0	0	0
5	1	0	0	0	0	0	0	0	0	0
6	1	0	0	0	2	0	0	0	0	0
7	1	0	0	0	2	0	0	0	0	0
8	1	0	0	0	2	0	0	0	0	0
9	1	0	0	2	0	0	0	0	0	0
10	1	0	0	0	0	0	0	0	0	0

Note: "1" means replacing Vessel A. "2" means replacing Vessel B.

Table B-7

The comparison of the optimal sequences in different plans.

[illegible]

Table B-5. We regress the payoff of replacing Vessel A at Year 1 and the discounted payoff of delaying replacement at Year 1. The equation is expressed as follows:

$$Y = -123,524 + 2.37X - 3.5 \times 10^{-6}X^2 \quad (36)$$

where Y is the values of Column 3 in Table B-5 and X is the values of Column 2 in Table B-5. Using (36), we obtain the expected discounted values for delaying replacement of Vessel A, which are shown in Column 4 of Table B-5. Thus, if we decide to replace Vessel A at Year 1, we can obtain the present payoffs (Column 2 of Table B-5) and the expected payoffs of the subsequent replacement (Column 6 of Table B-5). If we delay the replacement at Year 1, we have the expected deferral payoffs (Column 4 of Table B-5). In one simulation path, if the sum of the present payoff and the expected payoff of the subsequent replacement is greater than the expected deferral payoff, we choose to replace Vessel A in this path. The replacement decisions at Year 1 are shown in Column 7 of Table B-5. Using the same procedure, we obtain the replacement decisions for Vessel A with Year 1 to Year 3 in the sample paths. The results are shown from Column 2 to Column 4 of Table B-6. Based on Table B-6, we can obtain the probabilities for vessel replacement decisions at each year and the option value for Sequence 3.

By repeating the above procedures, we can obtain the option values for all 91 sequences. We find that Sequence 3 and Sequence 54 have the highest option values in Plan ① and Plan ②, respectively. Comparing these two sequences and Sequence 91 in Plan ③, we know that Sequence 91 is the best scheme. The replacement probabilities and the option values of these three sequences are summarized in Table B-7.

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